1.0 Problem statement

Employers are in wish to recruit developers who may obtain high job satisfaction throughout the period of employment such that they achieve and maintain high productivity. To facilitate the pre-screening process, the team hopes to exploit the key factors that impact job satisfaction to assist structuring pre-interview surveys. Compensation, workplace interpersonal relationships, workload, and related education and professional experience may serve as primal indicators of job satisfaction. [1] The team is intrigued how they individually and correlatively impact job satisfaction. Further, the team aims to develop a pre-screening job satisfaction prediction model that allows employers to input certain performance parameters taken from interview surveys and retrieve corresponding predicted job satisfaction. Methods of building and improving the models will be discussed in sections 3.0 and 5.0. The model will conduce to an efficient and effective job applicants screening and decision-making process and potentially offer career planning implications for developers.

2.0 Data

The raw dataset comes from a recent public survey[2] taken by the StackOverflow online developer community which includes 84 attributes and over 80,000 rows. It contains a comprehensive list of questions to the developers' career and personal life. We carefully screened feature attributes and dropped those out of scope, like *"Do you think this survey takes too long?"*. The final adopted features are briefly explained in the [Appendix 2]. Discarded features are available in [Appendix 1]. Overall, the EDA-ready data frame has one target variable: "Job Satisfaction" and around 50 feature variables. Some presumed key variables are *EdlLevel, YearsCode, ConvertedComp* [Appendix 2]*, etc.*

3.0 Methods

The team conducted Exploratory Data Analysis (EDA) in cleaning and transforming the raw dataset and visualizing certain statistical relationships which provided useful insights for model selections; findings are discussed in section 4.0. To predict job satisfaction, the team established a baseline model as a reference, logistic regression and random forest as two preliminary models to train and test the data.

4.0 Initial findings from EDA

- Data preprocessing starts with parsing strings[Appendix 3]. For multiple choice responses, semicolon-delimited strings were parsed and one-hot-encoded into binary variables. An example would be *"What programming language(s) do you work with?"* If a respondent selects *"C;Java;Python"*, then variables under columns C, Java and Python are set to 1, other unchecked languages are set to 0.

- In the handling of missing values, we dropped around 50,000 rows that contain at least one missing value since we have a fruitful dataset. We dropped another 3000 rows that contain inconclusive survey responses which cannot be represented by categorical variables, for instance, *"I don't know. / I prefer not to say."* are eliminated.[Appendix 4]

- Dealing with the target variable, we converted the likert-scale job satisfaction into numerical values. [Appendix 5] We ascendingly aligned 0 - 4 correspondingly with very dissatisfied - very satisfied. Same methodologies were applied to categorical variables that contain logical increments. For example, increasing education levels were converted into likert-type numeric values. For categorical variables that contain no logically incremental responses, like various indiscriminable undergrad majors, they are one-hot-encoded into binary variables.

- EDA starts with a sns bar-plot delineation of the distribution of job satisfactions. (*Figure 4.0*) It shows a left skewed graph. This sparks the idea of a baseline model which always predicts the mean job satisfaction from the training dataset. Next, we ran a test of correlation between every single feature variable and the target variable.[Appendix 6] Several paradigmatic positively related and negatively related correlations are further analyzed. The most correlated feature is *"Confidence in manager's capability"*. We created a proportional bar chart (*Figure 4.2*) showing the proportions of *"very*

confident in manager", *" Somewhat confident in manager"*, and *"Not at all confident"* regarding each rating of job satisfaction. It appears that the higher job satisfaction rating participants report, the more of them possess confidence in managers.

- Compensation is another evident indicator of job satisfaction. We plotted a scatter plot (*Figure 4.1*), showing the overall positive correlative trend between annual salary(*'ConvertedComp'*) and job satisfaction, while it further tells a curved nonlinear relationship which may indicate model preference on random forest and logistic regression rather than linear regression. The team attempted to find the job satisfaction distribution for all combinations of '*InfluComp*' and *'ConfiManager'* which are the two most correlated features to job satisfaction. The results are shown in [Appendix 7], indicating a combination of higher confidence in manager and higher influence in company leads to higher job satisfaction.

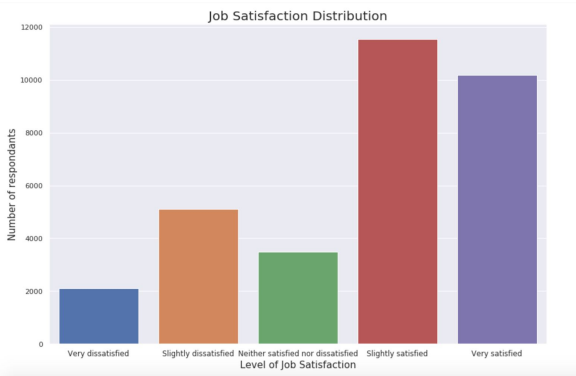


Figure 4.0. Bar-plot of job satisfaction

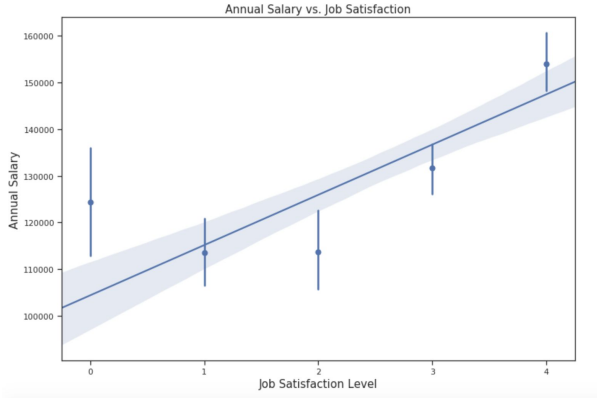


Figure 4.1. Scatter plot of trend between compensation and Job satisfaction

Figure 4.2. Stacked bar plot of confidence in manager

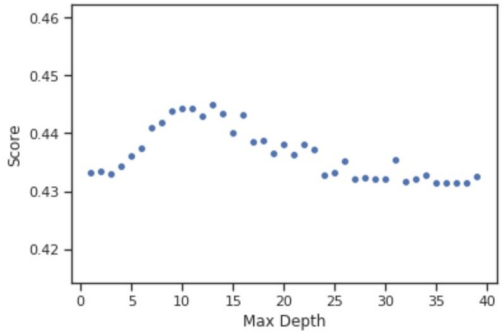


Figure 5.0. Scatter plot of Max_depth and score

5.0 Initial Findings form Current Model Building

A baseline model, a multiclass logistic regression model and a random forest classifier were selected as our initial set of models. Even though a decision tree could also achieve an accuracy of 0.43, its mean absolute error was as worse as the baseline model thus it's not selected. For each model below, we used a classification report to obtain precision score and recall score of each class as well as the accuracy for the model, and used confusion matrix for multiclass classification to evaluate the performance of classification models. In addition, the method of mean absolute error is employed to measure the average of the difference between the predicted value and the true value. A table of summary is presented below:

	Baseline Model	Logistic Regression Model	Random Forest Classifier
Accuracy	0.36	0.43	0.44
Mean Absolute Error (**MAE**)	0.943	0.876	0.865
Sum of Diagonal Value (**SDV**)	2311	2064	2051

Table 1. Overview of model performances

- With a left skewed distribution of job satisfaction (*Figure 4.0*), the probability that an applicant's job satisfaction is 3 or 4 is significantly high. A baseline model [Appendix 8] is employed to always predict job satisfaction to be the average job satisfaction of the training set: JobSat = 3. Since this model only predicts 3, the precision is 0.36 and the recall is 1.0 for the "JobSat = 3" class, while the recall and precision for all other classes are 0. The final model accuracy is 0.36. And SDV is 2311 in the confusion matrix. This is a benchmark for any future built model that will presumably perform better than the baseline model.

- The team established a multinomial logistic regression model[Appendix 9] that predicts the non-binary target variable. With similar methodologies applied, we acquired an accuracy of 0.43, which performs 7% better than the baseline model. The team applied a L1-regularization [Appendix 10] that automatically performs cross-validation to increase accuracy, while it rendered an arguably slight improvement of 0.003. The team also tried Ridge and Lasso regularization[Appendix 11] While Ridge is powerful in its wide range of feature variables considered and Lasso eliminates the unimportant features, they performed unexpectedly worse than the original regression model.

- The team further established a random forest classifier model[Appendix 12] for its quickness to train, excellent feature importance indications, and its great applicability to our 5-level target variable. A preliminary parameter tuning technique was applied by running multiple times of test with different maximum depths to find the locally optimal model. The team acquired the highest accuracy of 0.44 when maximum depth is set as 13(*Figure 5.0*). As a result, it performs slightly better than the logistic regression model,with a 1% increment. The SDV is 2051 in the confusion matrix, slightly less than that of regression model. Grid search was attempted to adjust the number of estimators, max_features, and several other parameters, unfortunately it failed to boost the accuracy of the model.

In conclusion, the performance of random forest classifier model is relatively the best through evaluating the output results across these three models. Through visualizations of 5*5 confusion matrices of these three models, the diagonal values of the confusion matrix for baseline model is relatively the largest, indicating more correct predictions.
From the correlation table we knew that a lot of features did not correlate well with our target. Significant improvement in correlation was observed in monotonic transformation of annual salary and interactions between categorical variables [Appendix 13]. However, final model performances were not improved.

6.0 Completion Plan
Even though grid search, cross validation, and feature engineering were attempted, the performances of regression and random forest models were not significantly enhanced. Some future plans to improve the model performance are:
Rather than removing rows with missing data and inconclusive responses, we will try to impute them. For example, we can add an extra category for the missing categorical variables. In addition, we might want to revisit the features that we originally regarded as uncorrelated. A stacking model may be developed in training the current available models and use them as additional features to improve the performance.
Seeing that MAE and accuracy are our main model performance indicators. If the actual job satisfaction is 4, while the predicted is 3, this would be a classification error. While for the fact that a higher-than-expected job satisfaction is what employers pursue, a model that predicts job satisfaction higher than the actual one might be the desirable one that we aim for.

Eventually, we want to develop a job application pre-screening simulator that maximizes average job satisfaction. It is a preliminary deterministic simulator. Inputs to this simulator would be applicant survey responses and certain criteria such as gender ratio; complementary technical skillset; number of senior and junior roles; and the volume of recruitment in the current hiring season.A what-if case simulator could be adopted for further system performance analysis as well, essentially tracking the change in performance as the inputs change.

7.0 Reference

[1] E. A. Locke, "What is job satisfaction?," *Organizational Behavior and Human Performance*, 26-Aug-2004. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0030507369900130. [Accessed: 31-Oct-2019].

[2] S. Overflow, "Stack Overflow 2018 Developer Survey," *Kaggle*, 15-May-2018. [Online]. Available:https://www.kaggle.com/stackoverflow/stack-overflow-2018-developer-survey#survey_results_public.csv
[Accessed: 31-Oct-2019].

8.0 Appendix

Appendix 1. Eliminated data dictionary

Eliminated feature	**Definition**
Respondent	Randomized respondent ID number
LastInt	What you were asked to do in the most recent successful job interview
FizzBuzz	1 if you have been asked to solve FizzBuzz in an interview, 0 if not
CurrencySymbol	Type of currency used in day-to-day
CurrencyDesc	Type of currency used in day-to-day
CompTotal	Total compensation of currency
CompFreq	Weekly, monthly, or yearly currency compensation
CodeRev	1 if review code as a work, 0 if not
CodeRevHrs	Number of hours on reviewing code per week
UnitTests	1 if company employs unit tests in the developing of products, 0 if not
PurchaseHow	How does your company make technology-related decisions
DevEnviron	Which development environment(s) do you use regularly
OpSys	What is the primary operating system in which you work?
Containers	How do you use containers
BlockchainOrg	How is your organization thinking about or implementing blockchain technology
OpenSourcer	How often do you contribute to open source?
MgrMoney	1 if you believe that you need to be a manager to make more money, 0 if not
BlockchainIs	What is Blockchain / cryptocurrency technology
BetterLife	1 if you think people born today will have a better life than their parents, 0 if not
ITperson	1 if you are the "IT support person" for your family, 0 if not
OffOn	1 if you have tried turning it off and on again, 0 if not
Extraversion	1 if you prefer online chat or IRL conversations, 0 if not
ScreenName	What do you call it
SOVisit1st	When did you first visit Stack Overflow
SOVisitFreq	How frequently would you say you visit Stack Overflow
SOVisitTo	The purpose of visiting Stack Overflow
SOFindAnswer	how many times a week do you find (and use) an answer on Stack Overflow
SOTimeSaved	What is your last time to solve a coding problem using Stack Overflow,
SOHowMuchTime	How much time did you save
SOAccount	1 if you have a Stack Overflow account, 0 if not
SOPartFreq	How frequently would you say you participate in Q&A on Stack Overflow
SOJobs	Have you ever used or visited Stack Overflow Jobs
OpenSource	How do you feel about the quality of open source software
MgrWant	1 if you want to become a manager yourself in the future, 0 if not

EntTeams	1 if you ever used Stack Overflow for Enterprise or Stack Overflow for Teams, 0 if not
SOComm	1 if you consider yourself a member of the Stack Overflow community, 0 if not
WelcomeChange	How welcome do you feel on Stack Overflow
SONewContent	1 if you like to see any of the following on Stack Overflow, 0 if not
Trans	1 if you identify as transgender, 0 if not
Sexuality	What sexual do you currently identify as
SurveyLength	How do you feel about the length of the survey this year
SurveyEase	How easy or difficult was this survey to complete
ResumeUpdate	What is the primary reason that you updated your resume
DatabaseWorkedWith	What database environments have you done extensive development work in over the past year, and which do you want to work in over the next year
PlatformWorkedWith	What platforms have you done extensive development work for over the past year
WebFrameWorkedWith	What web frameworks have you done extensive development work in over the past year, and which do you want to work in over the next year
EduOther	What types of non-degree education have you used or participated in
Student	1 if you currently enrolled in a formal, degree-granting college or university program, 0 if not
YearsCode	Number of years have you been coding
MiscTechWorkedWith	What other frameworks, libraries, and tools have you done extensive development work in over the past year, and which do you want to work in over the next year
Dependents	1 if you have any dependents (e.g., children, elders, or others) that you care for, 0 if not
SocialMedia	What social media site do you use the most
LanguageDesireNextYear	What programming, scripting, and markup languages have you done extensive development work in over the past year, and which do you want to work in over the next year
DatabaseDesireNextYear	What database environments have you done extensive development work in over the past year, and which do you want to work in over the next year
PlatformDesireNextYear	What platforms have you done extensive development work for over the past year
WebFrameDesireNextYear	What web frameworks have you done extensive development work in over the past year, and which do you want to work in over the next year
MiscTechDesireNextYear	What other frameworks, libraries, and tools have you done extensive development work in over the past year, and which do you want to work in over the next year
CareerSat	How satisfied are you with your career thus far
LastHireDate	When was the last time you took a job with a new employer
JobSeek	Which best describes your current job-seeking status
WorkRemote	How often do you work remotely
MainBranch	Which options best describes you today
WorkChallenge	what are your greatest challenges to productivity as a developer

Appendix 2. Adopted Data dictionary

Feature	**Definition**
Employment	current employment status
Country	country of residence
EdLevel	completed highest level of formal education
Orgsize	number of workers in the organization/company
Age1stCode	age of writing the first line of code
ImpSyn	rate of the level of competence
InfluComp	level of influence/power in the company when making technology-related decisions
Age	age (in years)
Gender	gender
BED	Developer back-end
FED	Developer front-end
Hobbyist	1 if coding is worker's hobby, 0 if not
YearsCodePro	number of years coding professionally
JobSat	level of satisfaction to current job
ConfiManager	Level of confidence in manager's capability
ConvertedComp	salary converted to annual USD salaries
WorkWeekHrs	number of working hours per week
Workplan	Level of working structure
WorkLoc	preferred places to work
Ethnicity	Descent
DBA	Data or business analyst
DSML	Data scientist or machine learning specialist
FSD	Full-stack developer
DE	Data engineer
OrgDiversity	Diversity of the company or organization
FlexSchedule	Diversity of the company or organization
OrgFinancial	Financial performance or funding status of the company or organization
Remote	Remote work options
WorkImpact	How widely used or impactful my work output would be
OfficeEnv	Office environment or company culture
ProDevelop	Opportunities for professional development
C	Programming language(1 if the respondent has this skill)
C#	Programming language(1 if the respondent has this skill)
Clojure	Programming language(1 if the respondent has this skill)
Elixir	Programming language(1 if the respondent has this skill)
C++	Programming language(1 if the respondent has this skill)
Kotlin	Programming language(1 if the respondent has this skill)
Scala	Programming language(1 if the respondent has this skill)

VBA	Programming language(1 if the respondent has this skill)
HTML/CSS	Programming language(1 if the respondent has this skill)
Java	Programming language(1 if the respondent has this skill)
JavaScript	Programming language(1 if the respondent has this skill)
Python	Programming language(1 if the respondent has this skill)
R	Programming language(1 if the respondent has this skill)
SQL	Programming language(1 if the respondent has this skill)
Computer Science/Computer Engineering	Field of study(1 if it is the major program for the respondent)
Information System	Field of study(1 if it is the major program for the respondent)
Natural Science	Field of study(1 if it is the major program for the respondent)
Other Engineering Disciplines	Field of study(1 if it is the major program for the respondent)
Web development/design	Field of study(1 if it is the major program for the respondent)
Business	Field of study(1 if it is the major program for the respondent)
Mathematics/Statistics	Field of study(1 if it is the major program for the respondent)
Fine Arts	Field of study(1 if it is the major program for the respondent)
Social Science	Field of study(1 if it is the major program for the respondent)
Humanities	Field of study(1 if it is the major program for the respondent)
Health Science	Field of study(1 if it is the major program for the respondent)

Appendix 3. Parsing Code

```
[ ] #-------------Parse_Step1.-------------------
    def Parse(val):
      counter_list =[]
      new = val.str.split(";",expand = False)
      for k in new:
            counter_list.append([item for i,item in enumerate(k)])
      return counter_list
```

```
[ ] df_copy = df.copy()
```

```
[ ] #-------------Parse_Step2.-------------------
    def ParsePlus(dev,types_interest):
      dfy = pd.DataFrame(columns = types_interest)
      for iter in dev:
        row_to_append = []
        for typ in types_interest:
            if typ in iter:
              row_to_append.append(1)
            else:
              row_to_append.append(0)
        s = pd.Series(row_to_append, index=dfy.columns)
        dfy = dfy.append(s, ignore_index=True)
      return dfy
```

```
[ ] #--------------Select options from multiple choice types attributes-------------------------
    # 1. DevType
    types_interested = ['Data or business analyst','Data scientist or machine learning specialist',
                        'Developer, back-end',
                        'Developer, front-end','Developer, full-stack','Engineer, data']

    # 2. JobFactor
    factors_interested = ['Diversity of the company or organization','Flex time or a flexible schedule',
                          'Financial performance or funding status of the company or organization',
                          'Remote work options','How widely used or impactful my work output would be',
                          'Office environment or company culture','Opportunities for professional development']

    # 3. LanguageWorkwith
    languages_interested = ['C','C#','Clojure','Elixir','C++','Kotlin','Scala','VBA','HTML/CSS','Java',
                            'JavaScript','Python','R','SQL']
```

```
[ ] #---------------Parsing by using 2 functions above----------------------
    # 1. Parse DevType
    devt = Parse(df_copy['DevType'])
    df_1 = ParsePlus(devt,types_interested)

    # 2. Parse JobFactor
    df_copy = df_copy.reset_index(drop=True)
    job = Parse(df_copy['JobFactors'])
    df_2 = ParsePlus(job,factors_interested)

    # 3. Parse LanguageWorkwith
    df_copy = df_copy.reset_index(drop=True)
    lang = Parse(df_copy['LanguageWorkedWith'])
    df_3 = ParsePlus(lang,languages_interested)
```

Appendix 4. Eliminated Non-Categorical Variables

```
[ ] df.drop(df.index[df['UndergradMajor']=='I never declared a major'],inplace = True)
    df.drop(df.index[df['MgrIdiot']=="I don't have a manager"],inplace = True)
    df = df.drop(df[df.Ethnicity.isin(["I don't know", "I prefer not to answer"])].index)
    df = df.drop(df[df.YearsCodePro.isin(["Less than 1 year","More than 50 years"])].index)
    df = df.drop(df[df.Gender.isin(["Man;Non-binary, genderqueer, or gender non-conforming",
                                    "Woman;Man","Woman;Man;Non-binary, genderqueer, or gender non-conforming",
                                    "Woman;Non-binary, genderqueer, or gender non-conforming",
                                    "Non-binary, genderqueer, or gender non-conforming"])].index)

    print(df.groupby(['Gender']).size())
    print(df.shape)
```

```
Gender
Man      30126
Woman     2317
dtype: int64
(32443, 22)
```

***'MgrIdiot' was later renamed as ConfiManager.*

Appendix 5. Converted the Likert-scale Job Satisfaction Question into 5 Categorical Variables

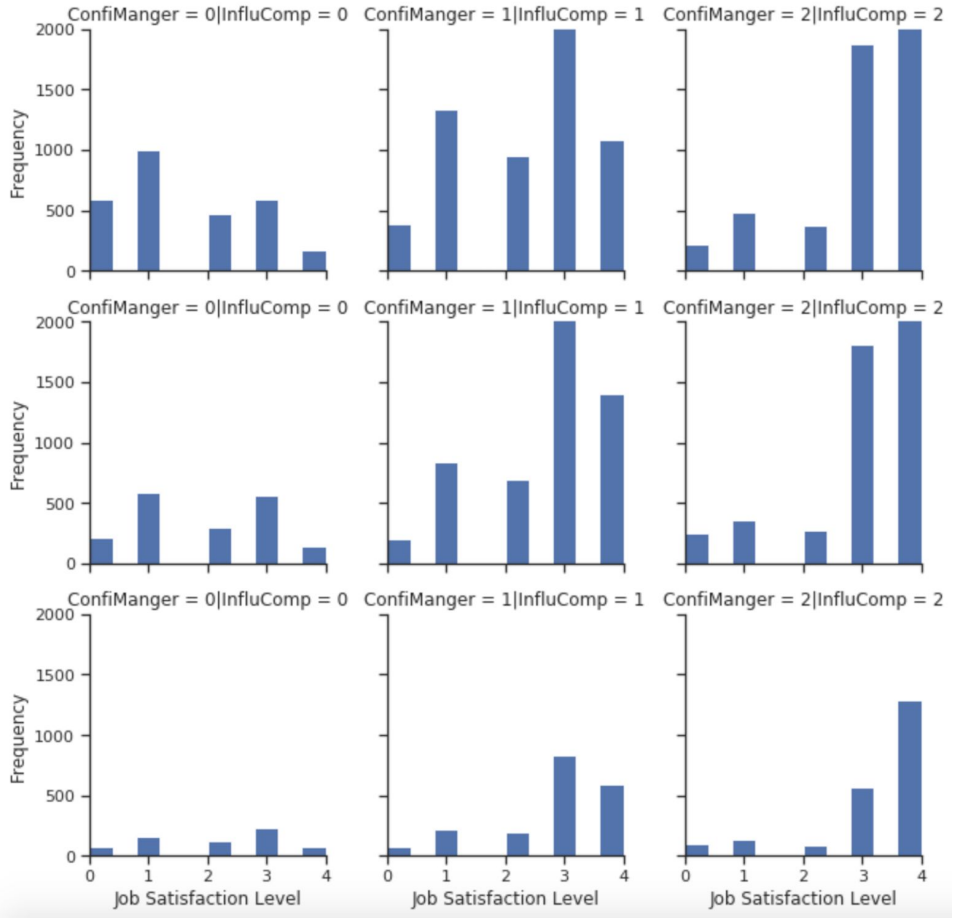
```
[ ] df_new['JobSat'].replace(['Very dissatisfied','Slightly dissatisfied',
                              'Neither satisfied nor dissatisfied','Slightly satisfied',
                              'Very satisfied'],[0,1,2,3,4],inplace=True)
```

Appendix 6. Correlation Output (Descending order):

JobSat	1.000000
ConfiManager	0.392543
InfuComp	0.152445
Office	0.111234
WorkPlan	0.088907
CompanyCulture	0.059883
ConvertedComp	0.045753
Python	0.032405
Woman	0.024092
Natural Science	0.022773
Scala	0.017774

Hobbyist	0.015997
DSML	0.015547
Humanities	0.014542
R	0.014077
Mathematics/Statistics	0.014027
HTML/CSS	0.011941
Social Science	0.011752
DE	0.010887
Fine Arts	0.010754
YearsCodePro	0.010634
Kotlin	0.010112
Health Science	0.009076
Elixir	0.007714
FED	0.006263
WorkImpact	0.006161
Clojure	0.006090
WorkWeekHrs	0.005306
JavaScript	0.004598
OrgDiversity	0.004024
FlexSchedule	0.003404
Business	0.003064
C++	-0.000587
Age	-0.001545
C	-0.003921
Computer Science/Computer Engineering	-0.005572
EdLevel	-0.006406
FSD	-0.006557
SQL	-0.008258
Other Engineering Disciplines	-0.009396
DBA	-0.012102
C#	-0.013422
BED	-0.014883
Information System	-0.016517
Java	-0.017152
OrgFinancial	-0.017713
VBA	-0.021394
Web development/design	-0.021544
Man	-0.024092
Age1stCode	-0.029314
Remote	-0.032997
ProDevelop	-0.033573
Other places	-0.072493
Home	-0.075196

Appendix 7. Facet Grid for job satisfaction level distributions across two categorical features: InfluComp and ConfiManager



Appendix 8. Baseline Model

```
# the baseline model is a dummy model that always predicts the average jobsat of the training dataset

def dummy_model(y_train,y_test):
  length = len(y_test)
  score = math.ceil(y_train.mean())
  scores = []
  for i in range(length):
    scores.append(score)
  Scores = pd.Series(scores)
  return Scores
```

	precision	recall	f1-score	support
JobSat=0	0.00	0.00	0.00	467
JobSat=1	0.00	0.00	0.00	1010
JobSat=2	0.00	0.00	0.00	689
JobSat=3	0.36	1.00	0.53	2311
JobSat=4	0.00	0.00	0.00	1992
accuracy			0.36	6469
macro avg	0.07	0.20	0.11	6469
weighted avg	0.13	0.36	0.19	6469

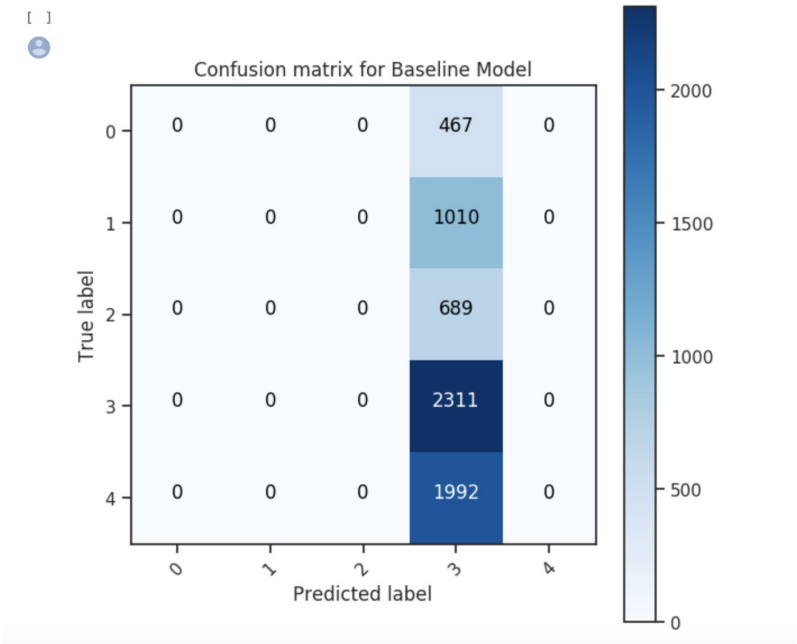


Figure. Confusion matrix for baseline model

Appendix 9. Logistic Regression model

```
logreg = LogisticRegression(multi_class='multinomial', solver='newton-cg')
logreg.fit(X_train, y_train)
```

```
              precision    recall  f1-score   support

     JobSat=0       0.00      0.00      0.00       467
     JobSat=1       0.35      0.10      0.15      1010
     JobSat=2       0.00      0.00      0.00       689
     JobSat=3       0.39      0.61      0.47      2311
     JobSat=4       0.50      0.63      0.55      1992

    accuracy                           0.43      6469
   macro avg       0.25      0.27      0.24      6469
weighted avg       0.35      0.43      0.36      6469
```

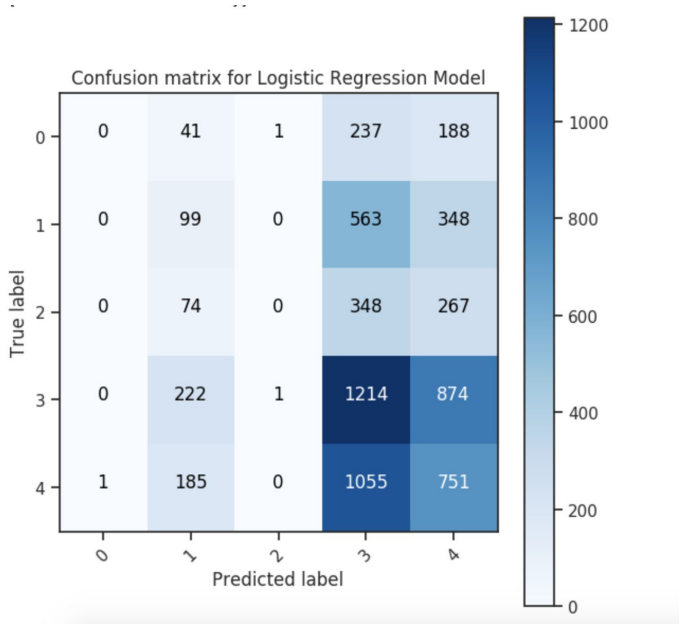


Figure. Confusion matrix for logistic regression model

Appendix 10. L1-Regularization

```
from sklearn.linear_model import LogisticRegressionCV

# Fit the model
logregcv = LogisticRegressionCV(penalty='l1', solver='saga')
logregcv.fit(X_train, y_train)

# calculate the score
train_score = logregcv.score(X_train, y_train)
test_score = logregcv.score(X_test, y_test)
print('The train score is {} and the test score is {}'.format(train_score, test_score))
```

Appendix 11. Lasso and Ridge Regularization

```
lasso = LassoCV()
lasso.fit(X_train, y_train)

# calculate the score
train_score = lasso.score(X_train, y_train)
test_score = lasso.score(X_test, y_test)
print('The train score is {} and the test score is {}'.format(train_score, test_score))

# calculate betas
betas = pd.Series(lasso.coef_, index=X_train.columns)
betas = betas.append(pd.Series({"Intercept": lasso.intercept_}))
print(betas)
```

Appendix 12. Random Forest Classifier model

```
[ ]
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)

    classifier = RandomForestClassifier(random_state = 0, max_depth = 13,
                                        n_estimators = 100, max_features = 0.2)
    classifier.fit(X_train, y_train)
```

```
              precision    recall  f1-score   support

     JobSat=0       0.22      0.00      0.01       467
     JobSat=1       0.32      0.20      0.24      1010
     JobSat=2       0.00      0.00      0.00       689
     JobSat=3       0.41      0.63      0.50      2311
     JobSat=4       0.53      0.60      0.56      1992

     accuracy                           0.44      6469
    macro avg       0.30      0.29      0.26      6469
weighted avg        0.38      0.44      0.39      6469
```

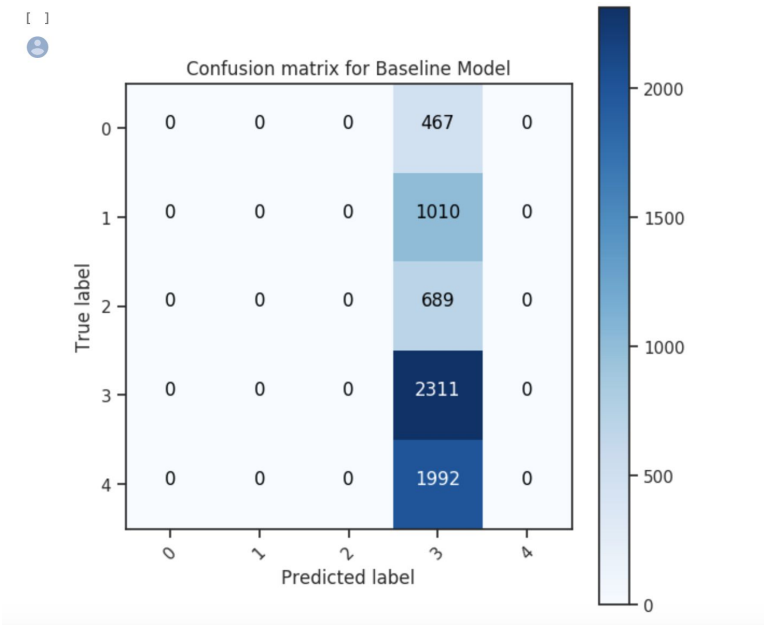


Figure. Confusion matrix for random forest classifier model

Appendix 13. Preliminary feature engineering

```
def new_feature_combos(X):
    X_new = X.copy()
    X_new['LogConvertedComp'] = np.log(X_new['ConvertedComp']+1)

    X_new['ImpSyn_**_Office'] = X_new.ImpSyn ** X_new.Office

    X_new['OtherPlaces_**_WorkPlan'] = X_new['Other places'] ** X_new.WorkPlan

    return X_new

X_train_interactions = new_feature_combos(X_train)
X_train_interactions.drop(columns=['ConvertedComp','ImpSyn','Office','Other places','WorkPlan'])
X_test_interactions = new_feature_combos(X_test)
X_test_interactions.drop(columns=['ConvertedComp','ImpSyn','Office','Other places','WorkPlan'])
```

```
Ridge = RidgeCV()
Ridge.fit(X_train, y_train)

# calculate the score
train_score = Ridge.score(X_train, y_train)
test_score = Ridge.score(X_test, y_test)
print('The train score is {} and the test score is {}'.format(train_score, test_score))

# calculate betas
betas = pd.Series(Ridge.coef_, index=X_train.columns)
betas = betas.append(pd.Series({"Intercept": Ridge.intercept_}))
print(betas)
```